Overnight Return, the Invisible Hand Behind Intraday Returns?

Ben Branch and Aixin Ma

An efficient market (weak form) will contain no significant price pattern, a view supported by numerous empirical studies. Our study, however, reveals a very strong negative autocorrelation between overnight and intraday returns, regardless of our sampling method or the methodology in use. Though this poses potential market mispricing opportunities, we conclude that future studies are needed in order to determine whether anyone other than a market maker can fully exploit these opportunities.

■In Fama's (1970) survey, almost none of the studies offer strong counterevidence to the "efficient markets" hypothesis for the stock market. Since then, however, studies of actual market performance reveal a number of apparent inconsistencies, including the January effect (Bhardwaj and Brooks, 1992; Thaler, 1987a; Thaler, 1987b), day-of-the-week effect (French, 1980; Kamara, 1997), other seasonal effects (Ariel, 1987; Cadsby and Ratner, 1992; Lakonishok and Smidt, 1988; Rozeff and Kinney, 1976; SEC Litigation Release 19033, 2005), size effect (Reinganum, 1981), value line enigma (Stickel, 1985), low price-to-earnings (PE) effect (Banz, 1981; Basu, 1977; Reinganum, 1981), and a variety of other anomalous security market behaviors such as value verses growth stock performance (La Porta, Lakonishok, Shliefer, and Vishny 1997; Agrawal and Tandon, 1994).

Our study of the daily returns for stocks listed on the NYSE, AMEX, and NASDAQ reveals a potentially anomalistic relationship that may have significant market implications. According to the weak form of the efficient market hypothesis, which is supported by numerous studies, a time series of returns should either be uncorrelated or the magnitude of any possible correlation should be too small to be financially meaningful. Our finding, that overnight stock returns are strongly autocorrelated with subsequent intraday returns, presents evidence that is inconsistent with the weak form of the efficient market hypothesis.

The current study was inspired by Branch, Ma, and Sawyer (2010), a study on how closed-end funds' returns were related to their net asset value (NAV) performance. Branch et al. (2010) bifurcate the daily returns of closed-end funds into overnight returns and intraday returns and find that the two exhibit a strong negative autocorrelation. The present study also expands the study by Branch and Ma (2008), which uses panel data analysis to uncover a strong autocorrelation between overnight returns and intraday returns for stocks listed on three major stock exchanges.

Our study confirms and extends in a more detailed way the finding of negative autocorrelation in stock returns as exhibited in Branch and Ma (2008) and Branch et al. (2010). We model our analysis in the mode of the Fama-MacBeth (1973) two-stage method. Our results are statistically significant and robust.

The remainder of this paper is organized as follows. We review the literature in Section I. Section II discusses the data sample and methodology. Section III reports results. Section IV discusses possible causes for autocorrelation in stock returns, including a brief discussion of literature on specialist behavior. Section V discusses practical implications of our findings, and Section VI concludes.

Ben Branch is a Professor of Finance in the Isenberg School of Management at the University of Massachusetts Amherst in Amherst, MA. Aixin Ma is an Associate Professor of Finance in the Meinders School of Business at Oklahoma City University in Oklahoma City, OK.

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I. Literature Review

A number of issues related to the relationship between overnight and intraday returns are considered in the market microstructure literature. For example, Hong and Wang (2000, p. 299) cite the following: "empirical patterns...a, intraday mean return and volatility are U-shaped; b, intraday trading volume is U-shaped; c, open-to-open returns are more volatile than close-to-close returns; d, weekend returns are lower than weekday returns; and e, returns over trading periods are more volatile than returns over non-trading periods."

Stoll and Whaley (1990) report that open-to-open returns are more volatile than are close-to-close returns. They note that this greater volatility implies that the correlation between the daytime and the following overnight returns exceeds that of the overnight and the following daytime returns, which in turn indicates that open-to-open returns are more likely to reverse than are close-to-close returns. They conclude that "the serial correlations results for the components of the 24-hour return indicate that the overnight return tends to be reversed by the following daytime return. This is an indication of a temporary price deviation at the opening. On the other hand, the daytime return is much less likely to be reversed by the return in the following night. This suggests that the closing price is less likely to reflect a temporary deviation" (p. 54). Curiously, they do not follow this line of inquiry to see where it might lead.

Pursuing another line of research, Harris (1989, p. 29) reports that "prices systematically rise near the end of the day and most obviously on the last trade…. The price rise is shown to be pervasive through time and across firms and strongest when the last transaction is near the close of trading." He goes on to report that "an increase in the frequency of ask prices at the day-end suggests that the last trade of the day is often initiated by a buyer" (p. 44). Inci, Lu, and Seyhun (2010, p. 323) discover that selling by corporate insiders tends to cause "contemporaneous stock price decline that is subsequently reversed."

Three working papers contribute to this literature. Kelly and Clark (2011, p. 132) report that "for a broad range of index exchange-traded funds (ETFs), the Sharpe Ratio of returns during non-trading hours ('close-to-open') exceeds the Sharpe Ratio of returns during trading hours ('open-to-close')." Somewhat similarly, Cliff, Cooper, and Gulen (2008, p. 23-24) report "…a surprising new pattern in returns; night returns, measured from close to open, are greater than day returns, measured from open to close. This pattern in returns produces the unexpected finding that the US equity premium (as measured by the S&P 500) over the last decade is solely due to overnight returns." Berkman, Koch, Tuttle, and Zhang (2009, 2011) further discover the high tendency for overnight returns to be positive and for intraday returns to be negative. They attribute the phenomenon to high open prices.

Our study seeks to add to this growing body of knowledge regarding nonrandomness in returns during the course of the day by adopting both parametric and nonparametric analyses, which do not assume any particular form of distribution for stock returns.

II. Sample and Method

Our study covers the time frame of January 1, 1994 to December 31, 2010. We used the Center for Research in Security Prices (CRSP) database for daily stock prices. To account for distortions caused by stock splits and dividends, we used the CRSP cumulative factor to adjust prices (CFACPR) to calculate adjusted close and open prices. We used the exchange traded Standard and Poor's Depositary Receipt (SPDR, with AMEX symbol SPY) as a proxy for the market instead of the S&P 500 for the following reason. The S&P 500 calculates its open price using component stocks' prices at 9:31 AM. Many stocks, however, are not opened by that time. As a result, the open price of the S&P 500 is often the sale price of its previous close, producing an S&P 500 0% overnight return for a vast number of the days. The SPDR, on the other hand, almost always opens by 9:31 AM with a current price based on the S&P 500 futures.

We categorize daily returns into three different groups – overnight returns, intraday returns, and close-to-close returns. The three groups of returns are calculated as follows.

$$overnight\ return_t = \left(\frac{open_t - close_{t-1}}{close_{t-1}} \right) * 100\%, \quad (1)$$

$$intraday\ return_t = \left(\frac{close_t - open_t}{open_t} \right) * 100\%, \quad (2)$$

$$close-to-close\ return_t = \left(\frac{close_t - close_{t-1}}{close_{t-1}} \right) * 100\%. \quad (3)$$

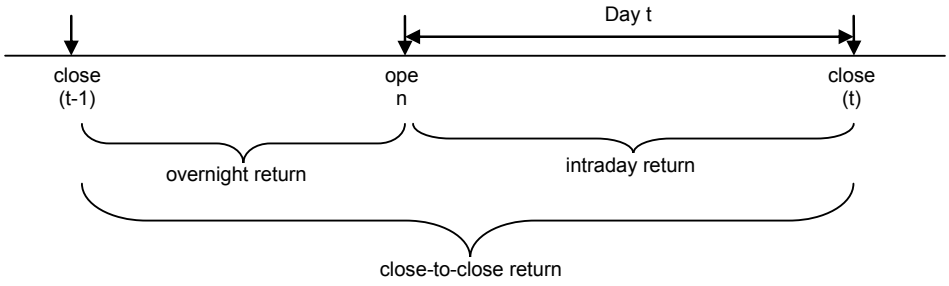
A visual representation of different returns is presented in Figure 1.

We model our analysis after the Fama-MacBeth two-stage method. In Stage One, we compute cross-sectional correlations on a daily basis and obtain a time series of daily correlation coefficients. In Stage Two, we calculate the means and perform statistical analyses using the time series result from Stage One.

For correlations involving various S&P 500 returns, we slightly adjust our method to address one issue: on any given day, returns for the S&P 500 are the same for all stocks, and all correlations associated with the S&P 500 returns

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Figure 1. Timeline of Returns



Note: Overnight, intraday and close-to-close returns are calculated as follows.

$$overnight\ \ return_t = \left(\frac{open_t - close_{t-1}}{close_{t-1}} \right) * 100\%;$$

$$intraday\ \ return_t = \left(\frac{close_t - open_t}{open_t} \right) * 100\%;$$

$$close - to - close\ \ return_t = \left(\frac{close_t - close_{t-1}}{close_{t-1}} \right) * 100\%.$$

appear to be zero. Our solution is straightforward. In Stage One, we calculate correlation coefficients using time series returns for each individual stock. In Stage Two, we calculate the means and perform statistical analysis by cross-section using the coefficients we obtained in Stage One.

We analyze the negative autocorrelation using two methods, the nonparametric univariate analysis and regression analysis. The univariate analysis presents at least one advantage over parametric analyses – it does not assume normal distribution of stock returns. In univariate analyses, we first calculate the means of intraday returns conditional on whether the associated overnight returns are up or down. We then tabulate these means of intraday returns based on the quartile of the overnight returns.

In the basic model of our regression analysis, we try to isolate the autocorrection among overnight returns, intraday returns, and the one-leg lagged overnight and intraday returns.

$$ID = \beta_0 + \beta_1 * ON + \beta_2 * lagID + \beta_3 * lagON + e, \qquad (4)$$

where the betas are various coefficients, *ID* and *ON* are the intraday and overnight returns of day *t*, and *lagID* and *lagON* are the intraday and overnight returns of day $t-1$. As a robust measure, we then proceed to run three more models adding the S&P 500, size, and various interaction terms as regressors.

Other robustness checks include analyzing the sample as a whole as well as analyzing different subsamples. For example, we isolated the two crisis periods of 2000 to 2001 and 2008 to 2009. We also divided our sample into pre- and post-decimalization periods. We analyze both unwinsorized and winsorized data by removing the top and bottom 0.05% of the observations. We analyze stocks on the NYSE, AMEX, and NASDAQ separately. To account for bid-ask bounce, we also substitute the closing bid-ask midpoint for the close price.[1]

III. Results

A quick scan of the correlation results reveals that both statistically and economically significant relationships exists between the overnight returns and the subsequent intraday returns (-.252). A similar relationship exists between lagged overnight and lagged intraday returns. Table I also suggests that S&P 500 returns could be a driving force behind stock returns because significant correlations also exist between overnight stock returns and overnight SPY returns (0.193) and between intraday stock returns and intraday SPY returns (0.187). Yet such a relationship between stock returns and S&P 500 returns cannot explain away the negative correlation between the overnight returns and intraday returns, a conclusion we draw after running regression analyses.[2]

Table II reports the means and quartiles of overnight and intraday returns for both the stocks and the SPY. For the study period, stocks in general have an upward bias as evidenced by the positive means and medians for both

[1] Robustness results are not reported but are available upon request.

[2] Results of regression analysis are reported in Table V.

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Table I. Correlation Matrix

ON, ID, SP_ON, and SP_ID stand for stock overnight return, stock intraday return, SPY overnight return, and SPY intraday return, respectively. LagXX stands for the lagged return for its respective stock return, and lagSP_XX stands for the lagged return for its respective SPY return.

Correlation coefficients are calculated in two stages. For ON, ID, lagON, and lagID, in Stage One we calculate correlation coefficients using cross-sectional data across all stocks on a daily basis and obtain a time series of correlation coefficients. In Stage Two, we calculate the means and perform statistical tests using the time series of correlation coefficients obtained in Stage One. Correlation coefficients among ON, ID, lagON, and lagID reported in Table I are such means calculated in Stage Two.

Correlation coefficients associated with SP_ON, SP_ID, lagSP_ON, and lagSP_ID cannot be calculated the same way because the S&P 500 return is the same for all stocks on any given day, causing all correlation coefficients to be zero for that day. For correlation coefficients associated with these S&P 500 return variables, in revised Stage One we calculate correlation coefficients using time series data for each stock over the entire study period and obtain stock-by-stock correlation coefficients. In revised Stage Two, we calculate the means and perform statistical tests using the stock-by-stock correlation coefficients obtained in revised Stage One. Correlation coefficients associated with SP_ON, SP_ID, lagSP_ON, and lagSP_ID reported in Table I are such means calculated in the revised Stage Two.

	ON	ID	SP_ON	SP_ID	lagON	lagID	lagSP_ON	lagSP_ID
ON	1.000	-0.252	0.193	0.006	0.028	-0.151	-0.015	0.036
ID		1.000	0.029	0.187	-0.039	0.015	0.000	0.014
SP_ON			1.000	-0.043	-0.021	0.012	-0.066	0.020
SP_ID				1.000	-0.006	-0.014	-0.009	-0.065
lagON					1.000	-0.244	0.185	0.006
lagID						1.000	0.028	0.184
lagSP_ON							1.000	-0.043
lagSP_ID								1.000

All correlation coefficients are significant at 0.01.

Table II. Distribution of Returns

Mean overnight and intraday returns for SPY are simple arithmetic means using daily SPY returns over the study period.

In Stage One, we calculate the mean overnight and intraday returns using cross-sectional data across all stocks on a daily basis and obtain a time series of daily means. In Stage Two, we calculate the mean, median, and other quartile returns and perform statistical tests using the time series of means we obtained in Stage One.

Students' *t*-values are in parentheses. Results are similar based on Sign test or Signed Rank test.

All calculations for SPY returns are based on unwinsorized data. All calculations for stock returns are based on winsorized data. We perform winsorization by first ranking the returns for all stocks and then removing the top 0.05% and bottom 0.05% of the ranked returns.

	Stocks		SPY	
	Overnight	Intraday	Overnight	Intraday
#Obs.**	4,280	4,280	4,280	4,280
Mean*	0.025% (3.390)	0.038% (2.870)	0.036% (3.570)	-0.004% (-0.260)
Max	5.101%	6.249%	6.068%	9.074%
75%	0.210%	0.444%	0.324%	0.524%
50%	0.034%	0.117%	0.061%	0.034%
25%	-0.141%	-0.332%	-0.227%	-0.513%
Min	-5.446%	-7.149%	-8.322%	-8.991%

**Number of observations reflects the number of days we calculated cross-sectional means in Stage One.

*Mean overnight and intraday returns for stocks are calculated in two stages.

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Table III. Distribution of Stock Returns – When Overnight Return is Down vs. When Overnight Return is Up

In Stage One, we first pool stocks into three groups – those with negative overnight returns, those with positive overnight returns, and those with overnight returns being zero. Then for each of the three groups, we calculate the mean overnight and intraday returns using cross-sectional data across all stocks in that group on a daily basis and obtain a time series of daily means for that group. In Stage Two, for each of the three groups we calculate the mean, median, and other quartile returns and perform statistical tests using the time series of means we obtained in Stage One for that group.

All calculations are based on winsorized data. We perform winsorization by first ranking the returns of all stocks and then removing the top 0.05% and bottom 0.05% of the ranked returns.

	Overnight Return is Down	Overnight Return is UP	Overnight Return is Zero
Mean* Overnight Return	-1.838% (-160.01)	1.914% (154.04)	0.000% (n/a)**
Mean* Intraday Return	0.658% (46.68)	-0.601 (-40.83)	0.005% (0.45)

**Because every overnight returns in this group is 0%, students' *t*-value cannot be calculated.
Students' *t*-values are in parentheses. Results are similar based on Sign test or Signed Rank test.
*Mean overnight and intraday stock returns are calculated in two stages.

Table IV. Detailed Distribution of Returns – When Overnight Return is Down vs. When Overnight Return is Up

In Stage One, we first pool stocks into three groups – those with negative overnight returns, those with positive overnight returns, and those with overnight returns being zero. Then within each group for each day, we rank the overnight returns and pool stocks into four quartiles based on the rank of their overnight returns for that day. Lastly we calculate the quartile mean return for each quartile of each of the three groups for each day using cross-sectional data across all stocks in that quartile for that day. As a result, we obtain a time series of quartile means. In Stage Two, we calculate the mean, median, and other quartile returns and perform statistical tests using the time series of quartile means we obtained in Stage One.

Students' *t*-values are in parentheses. Results are similar based on Sign test or Signed Rank test.

All calculations are based on winsorized data. We perform winsorization by first ranking the returns of all stocks and then removing the top 0.05% and bottom 0.05% of the ranked returns.

Overnight Return Percentile	0 – 25th	26 – 50th	51 – 75th	76 – 100th
Panel A. Overnight Return is Down				
Mean Overnight Return*	-4.765% (-170.97)	-1.539% (-141.41)	-0.749% (-132.51)	-0.299% (-110.93)
Mean Intraday Return*	1.828% (98.65)	0.528% (34.01)	0.226% (16.62)	0.062% (5.07)
Panel B. Overnight Return is Up				
Mean Overnight Return*	0.289% (108.01)	0.740% (130.46)	1.545% (137.15)	5.081% (162.94)
Mean Intraday Return*	-0.049% (-4.02)	-0.195% (-13.91)	-0.497% (-30.62)	-1.668% (-83.89)
*Panel C. Overnight Return is Zero***				
Mean Overnight Return	0.00% (N/A)			
Mean Intraday Return	0.039% (24.50)			

**Since every overnight return in this group is 0, all quartiles are the same and students' *t*-values cannot be calculated.
*Mean overnight and intraday returns of each quartile are calculated in two stages.

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intraday and overnight returns. No identifiable pattern is present in Table II except that the SPY has a positive mean for overnight returns but a negative mean for intraday returns. Since the negative mean for SPY intraday returns is not significant (t = -0.26), we should not read too much into the relationship. As soon as we bifurcate overnight returns depending on whether the return is up or down, a clear relationship between the overnight and intraday returns emerges.

Table III reports univariate results conditional on whether the overnight return is up or down. When the overnight return is down with a mean of -1.84%, the subsequent intraday return has a positive mean of 0.66%. On the other hand, when the overnight return is up with a mean of positive 1.91%, the subsequent intraday return has a mean of -0.60%. When the overnight return is zero, the subsequent intraday return has a mean that is statistically zero. All results are highly statistically significant.[3] The indication that overnight and intraday returns tend to move in opposite directions is clear. A further decomposition of overnight returns based on the magnitude of the return reveals that overnight and intraday returns not only tend to move in opposite directions, but also tend to move in a monotonic pattern – the further the overnight return moves in one direction, the further the intraday return tends to move in the other direction.

Table IV further decomposes the results in Table III into quartile groups based on the magnitude of overnight returns. Panel A reports overnight returns for each quartile and each quartile's associated intraday returns when the overnight return is down. The opposite directions for overnight and intraday price movement are unmistakable. The lowest overnight quartile, with a mean of -4.77%, has the highest intraday return of 1.85%. On the other hand, the highest overnight quartile, with a mean of -0.30%, has the lowest intraday return of positive 0.06%.[4]

Panel B reports overnight returns for each quartile and each quartile's associated intraday returns when overnight return is up. Once again, a strong, monotonic inverse relationship exists between overnight returns and intraday returns. For example, the highest overnight quartile, with a mean of positive 5.08%, has the lowest intraday return of -1.67%. The two panels together indicate that not only do positive (negative) overnight returns tend to be followed by negative (positive) intraday returns, but the more extreme the overnight movement, the greater the intraday reversal effect.

Table V reports multivariate regression results. Model 1, our basic model, suggests that a 1% overnight price movement will result in a 0.37% subsequent intraday movement in the opposite direction. This is consistent with results from Table IV. To check the robustness of our basic model, we also run Models 2, 3, and 4 by adding the S&P 500, size, and various interaction terms as regressors. Apparently adding more regressors does not dilute the explanatory power of the overnight coefficient. In fact, the overnight coefficient strengthens substantially to -0.89 in Model 2 and to -0.77 in Model 4. Nor does adding more regressors increase the overall explanatory power of our basic model, as indicated by the marginal improvement in adjusted R^2.[5] These robust results are consistent with what we discovered in our assumption-free nonparametric quartile analysis.

Another interesting observation is that the overnight return tends to have a higher carry-over effect for larger companies, as indicated by the positive interaction term of ON*CAP. This suggests that larger companies, which are generally more liquid, tend to experience fewer overnight-intraday price reversals. We believe that the observed size effect is linked to market-maker behavior, which we discuss in detail in the next section.

IV. What Causes the Negative Autocorrelation?

A. Market-Maker Behavior

Focusing on how the market makers (and most particularly specialists) are likely to open their assigned stocks can help us understand the causes for the negative autocorrelations described above.[6, 7]

The overnight return is based on the difference between the closing price and the opening price the following day. A stock will typically close somewhere within the range of its closing bid and ask. The closing price will contain some noise but does represent the market's real-time estimate of the stock's intrinsic value. If the bid-ask spread is relatively narrow, as is usual for an actively traded security, the close should be a rather reliable index of the market's view of the security's end-of-the-day worth. Even if the spread is wide, the close should generally be an unbiased estimate of the market's view.

Once the market closes, only those unfilled good-till-

[3] Even though Table III only reports t-values, results based on Sign test or Signed Rank test are similar.

[4] Panel A shows the group with the least negative overnight return.

[5] Note that all our results are highly significant; the high statistical significance is partially attributable to our enormous sample size.

[6] Our discussion herein uses the term "market maker" to refer to both the exchange specialist and the over-the-counter derivative (OTC) market maker since our data set includes both markets. We realize that the rules for the two types of market makers differ in certain respects. Our description is most appropriate for the specialist, but is also generally descriptive of OTC market makers as well.

[7] For a detailed analysis of the specialist system on the NYSE, see Stoll and Whaley (1990).

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Table V. Regression Analysis

Students' *t*-values are in parentheses. All returns are assumed to be normally distributed. ID, ON, and SP_ID stand for stock intraday return, stock overnight return, and SPY overnight return, respectively. LagXX is the lagged return for its respective stock return, and lagSP_XX is the lagged return for its respective SPY return. XXxCAP is the cross-term of ON (ID) × the natural log of the market capitalization, and lagXXxCAP is the lagged value of its respective XXxCAP term. We run Models 1 & 2 using the Fama-MacBeth two-stage regression method. In Stage One, we run one regression on all stocks for each day. In Stage Two, we calculate the mean and perform statistical tests using the time-series of regression coefficients obtained in Stage One. Because S&P 500 returns are the same for all stocks on any given day, running regression using the same Fama-MacBeth method would cause coefficients for all S&P 500 regressors to be zero. We therefore run Models 3 & 4 by pooling all returns into a combined panel dataset.

Model 1														**Adj. R^2**
ID =	β0	+	β1*ON	+	β2*LagID	+	β3*lagON							
	0.045%		-0.368		-0.035		-0.062							0.077
	(3.53)		(-280.19)		(-34.16)		(-62.67)							
Model 2														
ID =	β0	+	β1*ON	+	β2*LagID	+	β3*lagON	+	β4*ONxCAP	+	β5*LagIDxCAP	+	β6*LagONxCAP	
	0.052%		-0.890		-0.354		-0.236		0.029		0.018		0.010	0.085
	(71.16)		(-84.04)		(-50.15)		(-29.73)		(47.52)		(44.65)		(20.95)	
Model 3														
ID =	β0	+	β1*SP_ ON	+	β2*ON	+	β3*LagID	+	β4*lagON					
	0.041%		0.388		-0.357		-0.027		-0.055					0.063
	(58.21)		(362.04)		(-1381.50)		(-145.22)		(-212.29)					
Model 4														
ID =	β0	+	β1*SP_ ON	+	β2*ON	+	β3*LagID	+	β4*lagON	+	β4*ONxCAP	+	β5*LagIDxCAP	+ β6*LagONxCAP
	0.038%		0.355		-0.771		-0.390		-0.226		0.024		0.020	0.009 0.065
	(54.70)		(325.07)		(-319.16)		(-226.68)		(-94.75)		(169.54)		(211.15)	(68.52)

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cancelled (GTC) limit orders will remain. Any part of the quote provided by the market maker will also disappear at day's end. The absence of much of the previous day's unfilled orders is likely to widen the difference between the bid price (highest unfilled buy offer) and ask price (lowest unfilled sell offer) on the market maker's books. The wider the spread, the greater the market maker's latitude is in choosing where to open his or her stocks. An opening level most attractive to the market maker may be the most disadvantageous to many others. The market maker generally sets the opening between the highest unfilled bid and the lowest unfilled offer; but if the imbalance between buy and sell orders is big enough, the market maker may set the opening outside of this range.

For an actively traded security, orders both to buy and sell are likely to have entered the market prior to its opening. The orders received overnight, like the vast majority of orders received during the day, are mostly market (as opposed to limit, short, or stop) orders. At the beginning of the trading day, the buy and sell orders generally differ in share numbers; but all such market orders (as well as any consistent limit orders) must be executed at the opening.

When faced with an order imbalance, a market maker thus has incentives to set the opening either above or below the prior close and often overshoots the equilibrium price. As the day wears on, market price tends to revert to its intrinsic value, producing the observed negative autocorrelation.

The market maker could open the security at the previous close and make up the buy-sell imbalance out of inventory. Such a "leaning against the wind" strategy might be stabilizing but would have the market maker selling (buying) into a market with buying (selling) pressure, likely trading at a disadvantageous price against knowledge traders. Alternatively, the market maker could move the opening price toward the order imbalance. Thus if the imbalance involves an excess of sell (buy) orders, the security could be opened below (above) the previous close in order to trigger enough limit orders below (above) the prior close to offset the imbalance.

This approach has several advantages for the market maker. First, more limit orders are executed, and the market maker receives a fee for each transaction. Second, by using limit orders to cover part or all of the shortfall, the market maker limits changes in his or her inventory. Market makers usually have a target inventory level, some percent of the average daily trading volume for each security they handle. They tend to limit their exposure to unexpected overnight developments by ending each day with inventory close to that target level. Sometimes the market maker may end the trading day away from the target inventory level and therefore wish to adjust. Moving the opening price lower or higher can thus be more attractive to the market maker with a need to load or unload shares. Third, moving the price away from its prior close encourages trading in the market maker's assigned securities, thereby enhancing the opportunity for the market maker to earn more trading profits.

If no orders appear overnight, the market maker is unlikely to open the stock until new orders surface, sometimes well beyond the opening bell. When orders do surface, they are likely to be market orders that present the market maker with the same basic situation.

When faced with an order imbalance, a market maker thus has incentives to set the opening either above or below the prior close and often overshoots the equilibrium price. As the day wears on, market price tends to revert to its intrinsic value, producing the observed negative autocorrelation.

Market-maker behavior could also partially explain the observed size effect in our study. Larger companies tend to be more actively traded, have more orders at the open, and have more investors ready to take advantage of an imbalance on the open. Manipulating larger companies requires more capital commitment and attracts more attention; the smaller price movement commonly associated with these larger companies also provides less incentive to market makers. Therefore, we expect a lower price reversal during the day for larger companies when the price is more in line with the intrinsic value because of lower market-maker manipulation.

Real-world evidence such as Securities and Exchange Commission (SEC) rulings lends support to our reasoning that market-maker behavior could be a factor in producing the observed negative correlation between overnight and intraday returns.[8]

B. Literature on Specialist Behavior

How the specialist system works and how individual specialists behave receives a significant amount of attention, most of which is focused on NYSE specialists. Both Smidt (1971) and Barnea (1974) find differences in the ways individual specialist firms perform. Their studies exhibit variations in price volatility and bid-ask spreads across specialist firms even after adjusting for other factors. Corwin (1999) reaches a similar conclusion. In an extension of prior

[8] For one example, read Securities and Exchange Commission (SEC) Litigation Release No. 19033/January 13, 2005.

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work, Coughenour and Deli (2002) explore differences in the behavior of closely held verses widely owned specialist firms. Specifically, they find that those specialists whose owners are putting their own capital at risk have a greater incentive to limit adverse selection costs in their market-making function but are also more capital constrained in their ability to provide liquidity. As a result, owner-specialists are less inclined to stabilize the market at the opening than are other specialist firms. This finding implies that when faced with an order imbalance, specialists who put their own funds at risk are particularly inclined to allow their assigned stocks to open away from the prior close.

Bondarenko and Sung (2003) develop a model for specialist behavior in the presence of limit orders. They find that when the limit order book is deep, specialists have an incentive to trade with the market; when it is thin, they tend to be more inclined to trade against the market. In other words, when orders are imbalanced in a thin market, the specialist has an incentive to take the other side of the trade. Presumably that imbalance incentivizes the specialist to move the price temporarily in a direction in his/her favor. A subset of the literature, notably Cao, Choe, and Hatheway (1997) and Huang and Liu (2004), discovers that specialist firms use some of their market-making profits from actively traded securities to subsidize the market-making function of less actively traded securities.

C. Bid-Ask Bounce

Bid-ask bounce offers another possible explanation. Consider a stock that trades within the same bid and ask over several days. Suppose that on a particular day the stock closed at the bid. On the following day, the stock opens either at the bid or the ask. Opening at the bid results in no price change, but opening at the ask registers a positive overnight move even though the underlying quote has not changed. We observe the same phenomenon when the market closes again. So as long as the bid and the ask don't change, every price change will happen in the opposite direction from the preceding price change. In other words, the price will bounce back and forth between the bid and the ask, producing the observed negative autocorrelation even though the price has not moved at all. Furthermore, even when the bid-ask quotes do change, stocks randomly opening and closing at the bid and ask exhibit some degree of negative autocorrelation.

Lease, Masulis, and Page (1991) explore the impact of the bid-ask bounce, finding that the midpoint between the bid and ask is a better proxy for the "price" when an order imbalance is present. In a more recent study, Johnson, Johnson, and Shanthikuma (2008) explore the role of bid-ask bounce in stock returns and find that stocks that close just above a round number, e.g., $15.01, are considerably more likely to rise on the following day than are those that close just below a round number, e.g., $14.99.

Price "movement" caused by bid-ask bounce can be effectively dealt with using a midpoint quote between the bid price and the ask price, as suggested by Berkman et al. (2011). Bid-ask bounce does not seem to explain much of the observed microstructure anomalies, however. In our study, replacing the close price with the midpoint of the bid-ask spread has no meaningful impact on the results. In Johnson et al. (2008), even though removing the impact of bid-ask bounce reduces the magnitude by about 50%, it leaves the results robust. Additionally, a random bid-ask bounce does not produce overwhelmingly positive overnight returns versus overwhelmingly negative intraday returns, as observed by Berkman et al. (2011).

D. Alternative Explanations

Other researchers provide alternative explanations for the causes of the overnight-intraday reversal phenomenon. Berkman et al. (2011, p. 26) report that the phenomenon can be attributed to higher open prices and subsequent lower close prices, particularly "for stocks that are difficult to value and costly to arbitrage…," and is " …larger in magnitude during periods of high investor sentiment." Barber and Odean (2008) hypothesize that higher open prices are caused by retail investors' herding to buy attention-seeking stocks at the open. Whether such herd mentality also causes the same reversal effects in other securities as observed in Branch et al. (2010) and Cliff et al. (2008) has yet to be studied. Our future study will expand into statistical analysis of market-maker behavior and other possible macro and micro factors behind the overnight-intraday price reversal phenomenon.

V. Practical Implications

Can market participants formulate trading rules to take advantage of the overnight-intraday price reversal movement? One possible strategy would be to look for stocks that open away from their prior close; that is, buy a stock when it opens down or sell it when it opens up and then reverse the trade at the subsequent close. Implementing such a strategy, however, requires knowing which way to trade at the opening. To do so, one would need to know the direction of the imbalance of overnight orders. The market maker has such knowledge, but the public does not. Moreover, entering an order to be executed at the opening could itself impact the relationship. If, for example, a market order to buy was entered prior to the opening on the expectation that the stock will open down and rise over the course of the day, the very act of entering that buy order might reverse the selling imbalance and thus cause the stock to open up. Consequently, trying to trade effectively at the opening is

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likely to be difficult.

An alternative strategy would be to trade shortly after the opening. For this strategy to be effective, the stock would need to remain available for a trade near the opening level for some period after the opening. This again can prove to be very difficult. Transaction costs and slight price movement in the reverse direction will almost certainly wipe out any profit potential.

One can, however, avoid certain disadvantageous trading behaviors around the open. Specifically, entering a market order to be executed at the open is almost always inadvisable. Sometimes one could land on the weak side of the opening imbalance by luck. However, more often one would be on the side to be exploited.

VI. Conclusion

Using a large sample that spans from January 1, 1994 to December 31, 2010, we explore how overnight and intraday returns of US stocks are related. Our study reveals a very substantial degree of negative autocorrelation between adjacent overnight and intraday returns. The result is independent of sampling method and robust to different methodologies. Though this is inconsistent with the weak form of the efficient market hypothesis and creates a potential market mispricing opportunity, we conclude that anyone other than a market maker would have difficulty exploiting such a relationship.■

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